MULTIPLICITY VIA KOSZUL TOR FOR IDEALS GENERATED BY REGULAR SEQUENCES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original remark 0B00: sources/stacks-chow/tags.txt, source lines 13843–13843. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09CC: sources/stacks-chow/tags.txt, source lines 11825–11825. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original theorem 0AZZ: sources/stacks-chow/tags.txt, source lines 13842–13842. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let A be a Noetherian local ring. Let $I = (f_1, \ldots, f_r)$ be an ideal generated by a regular sequence. Let M be a finite A-module. Assume that* $\dim(\mathit{Supp}(M/IM)) = 0$. *Then*

$$e_I(M, r) = \sum (-1)^i \mathit{length}_A(\mathit{Tor}_i^A(A/I, M))$$

Here $e_I(M, r)$ is as in Remark 0B00.

Proof. Since $f_1, \ldots, f_r$ is a regular sequence the Koszul complex $K_\bullet(f_1, \ldots, f_r)$ is a resolution of A/I over A, see More on Algebra, Lemma 09CC. Thus the right hand side is equal to

$$\sum (-1)^i \text{length}_A H_i(K_\bullet(f_1, \ldots, f_r) \otimes_A M)$$

Now the result follows immediately from Theorem 0AZZ if I is an ideal of definition. In general, we replace A by $\overline{A} = A/\text{Ann}(M)$ and $f_1, \ldots, f_r$ by $\overline{f}_1, \ldots, \overline{f}_r$ which is allowed because

$$K_\bullet(f_1, \ldots, f_r) \otimes_A M = K_\bullet(\overline{f}_1, \ldots, \overline{f}_r) \otimes_{\overline{A}} M$$

Since $e_I(M, r) = e_{\overline{I}}(M, r)$ where $\overline{I} = (\overline{f}_1, \ldots, \overline{f}_r) \subset \overline{A}$ is an ideal of definition the result follows from Theorem 0AZZ in this case as well. □

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